



Differential effects of follicle-stimulating hormone dosage on in vitro embryo production in high- and low-anti-Müllerian hormone Holstein heifers

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ABSTRACT

Female cattle with low-anti-Müllerian hormone (AMH) have fewer antral follicles and reduced in vitro embryo production (IVEP) potential than high-AMH females. In addition, granulosa cells of low-AMH females have a diminished responsiveness to FSH, despite elevated circulating FSH concentrations. However, AMH phenotype is generally not considered when establishing FSH dosage for ovarian superstimulation. Thus, this study aimed to assess the effect of FSH dosage on IVEP outcomes in high- and low-AMH female cattle. Pregnant Holstein heifers (~50 d of gestation) with high (>331 pg/mL; n = 31) and low (<192 pg/mL; n = 31) circulating AMH were enrolled at 2 different locations for experiment 1 and experiment 2, respectively. Heifers were assigned to receive 0, 280, or 525 IU of FSH in a crossover design at 14 d intervals. Follicular wave emergence was synchronized using follicular ablation and FSH treatments, comprising 4 decreasing doses administered 12 h apart, were initiated 36 h after follicular ablation. Cumulus oocyte complexes (COC) were retrieved using ovum pick-up (OPU) 40 h after the last FSH administration and subjected to IVEP. Data were analyzed using generalized linear mixed models and orthogonal polynomial contrasts. In high-AMH heifers, total follicles and COC increased in a linear FSH dose-dependent manner. Similarly, blastocyst percentage increased linearly with increasing FSH dose, resulting in a linear increase in blastocyst yield (0 IU: 2.8 ± 0.4 ; 280 IU: 4.9 ± 0.7 ; 525 IU: 6.5 ± 0.8). In low-AMH heifers, there was a quadratic effect of FSH dose on total follicles, whereby there was a greater increase in number of follicles between 0 and 280 IU of FSH than between 280

and 525 IU. Conversely, the number of COC increased linearly with increasing FSH dose. There was a quadratic effect of FSH dose on blastocyst percentage, whereby the blastocyst percentage increased between 0 and 280 IU of FSH but not between 280 and 525 IU. Nevertheless, the number of blastocysts produced increased linearly with FSH dose (0 IU: 2.0 ± 0.4 ; 280 IU: 3.8 ± 0.6 ; 525 IU: 4.8 ± 0.8). In conclusion, the administration of FSH before OPU enhances IVEP in both high- and low-AMH heifers in a dose-dependent manner. Notably, the FSH-driven improvement in oocyte developmental competence reached a threshold at 280 IU of FSH in low- but not high-AMH heifers, suggesting that the optimal FSH dosage to maximize IVEP efficacy may be less for low- than high-AMH phenotype heifers.

Key words: FSH dose, bovine, in vitro embryo production, superstimulation, developmental competence

INTRODUCTION

The utilization of ovarian superstimulation regimens before ovum pick-up (OPU) and in vitro embryo production (IVEP) has become widespread in the bovine embryo transfer (ET) industry, such that ovarian superstimulation is implemented in ~61% and ~93% of OPU/IVEP procedures performed in North America and Europe, respectively (Viana, 2024). Follicular development in cattle exhibits a wave-like pattern (Pierson and Ginther, 1988; Sirois and Fortune, 1988), with the emergence of each follicular wave being stimulated by a surge in circulating FSH concentrations (Adams et al., 1992). The increase in FSH stimulates the growth of the recruited follicles, however, as the follicular wave develops, circulating FSH concentrations decline, reaching nadir levels near the time of deviation (Ginther et al., 1998), when a single follicle is selected, acquiring a dominant phenotype. Administration of FSH, during ovarian superstimulation, has been shown to rescue subordinate follicles from undergoing atresia rather than recruiting

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new follicles into the follicular wave, allowing multiple follicles to acquire a dominant phenotype (Adams et al., 1993; García Guerra et al., 2015). Interestingly, multiple studies have reported a greater number of follicles at OPU when ovarian superstimulatory regimens are used (reviewed by Motta et al., 2024). This unexpected observation may be attributed to improved follicle visualization as a result of increased follicle size or to the rescue of small (<2 mm) antral follicles, or both (Jaiswal et al., 2004; García Guerra et al., 2015). Considering the lesser antral follicle population in *Bos taurus* compared with *Bos indicus* breeds (Baldrighi et al., 2014; Batista et al., 2014; Sartori et al., 2016), this increase in follicle count at OPU is particularly advantageous when conducting OPU/IVEP in *Bos taurus* females. In addition, ovarian superstimulatory treatments enhance oocyte developmental competence, leading to relative increases of 60% to 380% in the number of in vitro produced embryos (reviewed by Motta et al., 2024). The use of ovarian superstimulation, however, adds substantial cost to OPU/IVEP procedures, emphasizing the need for more judicious FSH dosing to optimize efficiency.

Efficient utilization of superstimulation regimens requires the determination of optimal FSH dosages. Assessment of the optimal FSH dosage during superstimulation for multiple ovulation and ET (MOET) indicated a dose-dependent increase in ovarian response (Gonzalez et al., 1990; Souza et al., 2007; Krause et al., 2020) and transferable embryos (Gonzalez et al., 1990). However, doses exceeding the equivalent of 400 mg of National Institute of Health porcine FSH standard (NIH-FSH-P1) did not result in further increases in ovarian response or embryo development rates, suggesting the presence of a superstimulatory threshold (Gonzalez et al., 1990). In contrast, lower FSH doses ranging from 105 to 525 IU (60 to 300 mg of NIH-FSH-P1) are typically employed for OPU/IVEP (Vieira et al., 2014; Cedeño et al., 2022; Hayden et al., 2023), albeit research examining the effect of FSH dose on OPU/IVEP performance is relatively scarce. The possibility of a dose-dependent FSH effect on OPU/IVEP outcomes was first suggested by Meintjes et al. (1995) through the evaluation of different doses of a now discontinued FSH pituitary extract (i.e., FSH-P) in pregnant beef cows. Increasing the FSH-P dose by 2-fold was observed to increase total number of follicles, retrieved cumulus oocyte complexes (COC), and cleavage percentage, though statistical significance was likely hindered by the small sample size. More recently, our research group evaluated this dose-dependent effect in pregnant Holstein heifers using the commercially available porcine FSH extract, Folltropin (Hayden et al., 2021, 2023). Administration of 0, 280, or 525 IU of Folltropin (equivalent to 0, 160, or 300 mg of NIH-FSH-P1) to pregnant heifers revealed that the number of follicles, COC retrieved, and

viable COC increased linearly with increasing FSH dose (Hayden et al., 2023). Moreover, oocyte developmental competence and blastocyst yield followed a similar linear dose-dependent increase, whereby utilization of a 525-IU dose yielded 120% more blastocysts compared with the 0-IU dose. In agreement with these findings, results from another study indicate that increasing the dose of FSH from 280 to 525 IU resulted in a significantly greater number of visualized follicles, COC, and cleavage percentage, leading to a ~33% increase in the number of blastocysts produced (Hayden et al., 2021). These studies, however, did not consider phenotypic variability in factors such as antral follicle count (AFC), which may influence the response to ovarian superstimulation, underscoring, therefore, the need to evaluate FSH dosage in the context of phenotypic differences.

Variability in embryo production among females has been primarily attributed to differences in antral follicle population, commonly assessed through AFC or circulating anti-Müllerian hormone (AMH), or both. Females with relatively greater AFC/AMH present greater ovarian response to FSH and enhanced embryo production following both MOET (Ireland et al., 2007; Silva-Santos et al., 2014; Souza et al., 2015) and OPU/IVEP procedures (Batista et al., 2016; Motta et al., 2024; García-Guerra et al., 2025). The observed variability in AFC/AMH phenotype is accompanied by marked physiological differences. For instance, compared with high-AFC/AMH, low-AFC/AMH females exhibit several notable differences, including lower circulating androgen concentrations and reduced thecal cell responsiveness to LH (Mossa et al., 2010a); reduced luteal cell capacity for progesterone production and reduced circulating progesterone (Jimenez-Krassel et al., 2009); greater abundance of cytochrome P450 family 19 subfamily A member 1 (*CYP19A1*) mRNA in granulosa cells and greater estradiol concentrations in the follicular fluid (Ireland et al., 2009); and cumulus cells with greater abundance of cathepsin B (*CTSB*) and S (*CTSS*), as well as estrogen receptor 1 (*ESR1*) and 2 (*ESR2*; Ireland et al., 2009). More importantly, low-AFC/AMH cattle consistently have elevated circulating FSH concentrations (Singh et al., 2004; Burns et al., 2005; Ireland et al., 2007; Jimenez-Krassel et al., 2009; Mossa et al., 2010b), and their follicles contain granulosa cells with fewer FSH receptors and diminished capability to respond to FSH stimulation, as indicated by reduced *CYP19A1* mRNA abundance and estradiol production in response to FSH treatment (Scheetz et al., 2012). These findings collectively suggest that females with a low-AFC/AMH phenotype may require lesser FSH dosage than their high-AFC/AMH counterparts.

Accordingly, a series of recent experiments (Karl et al., 2021, 2023; Clark et al., 2022a,b) evaluated the ef-

fects of excessive FSH dosage (1,680 IU as opposed to 560 IU) during superstimulation in low-AFC phenotype females. Utilization of a high-FSH dosage led to: (1) reduced estradiol production; (2) decreased ovulation rate; (3) ovulatory follicles with follicular fluid with lesser estradiol and greater progesterone and oxytocin concentrations suggestive of premature luteinization; (4) dysregulation of maturational processes, including those involved in cumulus expansion and regulation of meiotic resumption; and (5) induction of expanded COC that prematurely resumed meiosis and responded poorly to IVEP. The FSH dose used in these studies, however, was ~3-fold greater than the largest dose currently used in superstimulatory treatment regimens for commercial OPU/IVEP procedures. Interestingly, assessment of the effect of clinically relevant FSH doses (i.e., 245 to 525 IU; Demetrio et al., 2020; García-Guerra et al., 2025) in the context of AFC/AMH phenotypes has yet to be fully assessed. The objective of the present study, therefore, was to evaluate the effect of FSH dosage on ovarian response and IVEP in heifers with low- and high-AMH phenotypes. It was hypothesized that FSH would enhance superstimulatory response, oocyte developmental competence, and in vitro embryo production in a dose-dependent manner in both low- and high-AMH heifers; however, maximal response would be attained at a lower FSH dosage for low- than high-AMH heifers. Phenotype-driven optimization of FSH dosage not only offers the potential to improve IVEP outcomes but also considerable economic advantages. By reducing FSH use in low-AMH donors and increasing it in high-AMH donors, this strategy can improve resource utilization and reduce per-embryo production costs, thereby enhancing the overall cost-effectiveness of IVEP programs.

MATERIALS AND METHODS

All animal procedures were approved by the Institutional Animal Care and Use Committee of the Office of Research at The Ohio State University (no. 2020A00000056) and were performed in accordance with the Guide for the Care and Use of Agricultural Animals. Two experiments were conducted at 2 facilities belonging to the same commercial IVEP program, located in Wisconsin (experiment 1: 43°29'24"N, 88°18'14"W) and Ohio (experiment 2: 39°49'22"N, 83°35'54"W). Pregnant Holstein heifers (50.3 ± 3.8 d of gestation) with circulating AMH concentrations in the upper (high AMH; >331 pg/mL; experiment 1) and lower (low AMH; <192 pg/mL; experiment 2) tertiles of a reference population were selected. Threshold AMH concentrations were derived from a large reference population composed of Holstein heifers housed at the same locations (Lemos Motta et al., 2021). Mean circulating AMH concentrations for the

reference population were 309.9 ± 221.9 pg/mL (range 8.3–1,913 pg/mL). Pregnant heifers were selected for this study due to greater oocyte competence and embryo yield as compared with nonpregnant heifers and because the ability to produce embryos from pregnant females is one of the main advantages of OPU/IVEP systems (Baruselli et al., 2016; García-Guerra et al., 2025). Heifers were housed in a compost-bedded pack confinement barn with free access to water and were fed twice daily a TMR formulated to meet or exceed the nutritional requirements for heifers weighing 350 kg (NASEM, 2021).

Experiment 1: High-AMH Phenotype Heifers

Experiment 1 evaluated the effects of FSH dosage on OPU/IVEP outcomes in high-AMH phenotype heifers. Pregnant Holstein heifers (n = 31), 17.7 ± 2.0 mo old, with BCS of 3.3 ± 0.3 on a 1 to 5 scale (Wildman et al., 1982), and circulating AMH concentrations of 627.9 ± 213.3 pg/mL (range 342.3–1,181.9 pg/mL) were assigned to a 3-treatment crossover design study conducted in 4 cohorts (6 to 9 heifers per cohort). Treatments consisted of the administration of porcine FSH (Folltropin, Vetoquinol USA, Ft. Worth, TX) at a total dose of 0 IU, 280 IU (equivalent to 160 mg of NIH-FSH-P1), or 525 IU (equivalent to 300 mg of NIH-FSH-P1) before OPU. Heifers within each cohort were randomly assigned to 1 of 6 possible treatment sequences. The 3 treatment periods for each heifer were initiated at 49 (range 47–51), 63 (range 61–65), and 77 (range 75–79) days of gestation, with a 14-d washout interval between treatment periods. Follicular wave synchronization was performed on d –4.5 (Figure 1) of each treatment period using follicular ablation of all follicles >5 mm, as previously described (Hayden et al., 2022). Follicle-stimulating hormone administration was initiated 36 h after follicular ablation (d –3) for the FSH280 and FSH525 groups and consisted of 4 decreasing i.m. administrations 12 h apart (Figure 1). On d 0, ~40 h after the last FSH administration, ovarian response was evaluated using a B-mode portable ultrasound scanning device (Ibex Evo II; E.I. Medical Imaging, Loveland, CO) equipped with a convex array transducer (9 MHz; Ibex Evo II; E.I. Medical Imaging, Loveland, CO) fitted into a plastic OPU handle. Follicles were measured using built-in calipers, counted, and assigned to the following predefined classes: small (<6 mm), medium (6–10 mm), or large (>10 mm).

Experiment 2: Low-AMH Phenotype Heifers

Experiment 2 evaluated the effects of FSH dosage on OPU/IVEP outcomes in low-AMH phenotype heifers. Pregnant Holstein heifers (n = 31), 17.8 ± 2.3 mo old, with a BCS of 3.7 ± 0.3 and circulating AMH concentra-

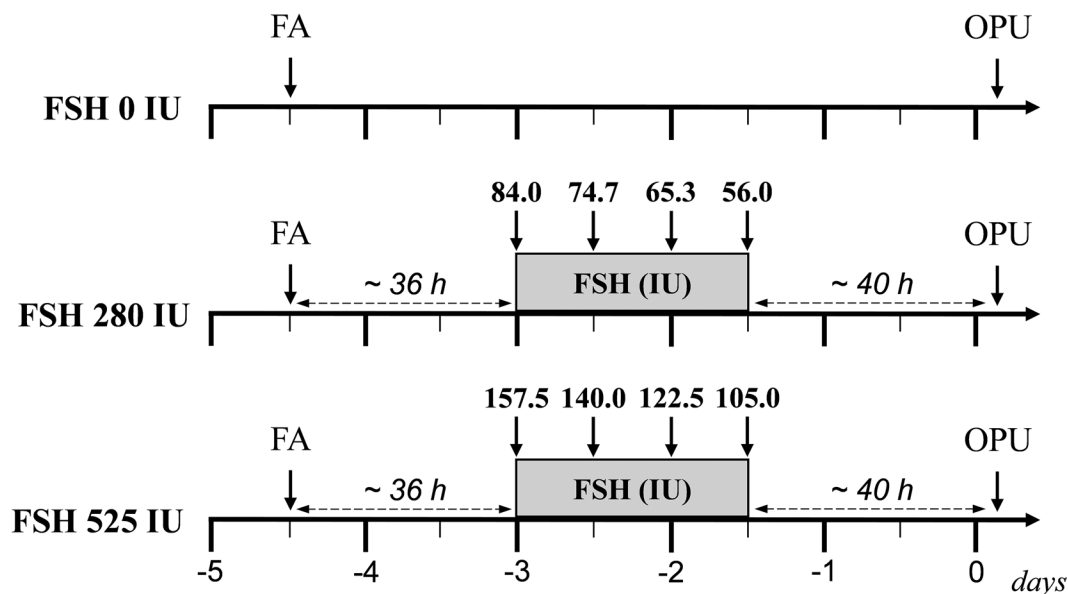


Figure 1. Schematic representation of treatments for experiments 1 and 2. High- (experiment 1) and low-AMH (experiment 2) pregnant Holstein heifers were administered 0 IU, 280 IU (equivalent to 160 mg of NIH-FSH-P1), or 525 IU (equivalent to 300 mg of NIH-FSH-P1) of FSH after synchronization of follicular wave emergence using follicle ablation (FA), and before ovum pick-up (OPU) and in vitro embryo production. Transvaginal ultrasonography (US) was performed at the time of OPU to determine the number and size class of ovarian follicles.

tions of 133.4 ± 41.8 pg/mL (range 17.2–189.5), were assigned to a 3-treatment crossover design study conducted in 3 cohorts (9 to 12 heifers per cohort). Treatments consisted of the administration of FSH at total doses of 0 IU, 280 IU, or 525 IU before OPU (Figure 1), and heifers within each cohort were randomly assigned to 1 of 6 possible treatment sequences. The 3 treatment periods for each heifer were initiated at 47 (range 36–57), 61 (range 50–71), and 75 (range 64–85) d of gestation, with a 14-d washout interval between treatment periods. Follicular wave synchronization, FSH administration, and the assessment of ovarian response were performed as described in experiment 1.

Ultrasound-Guided Ovum Pick-Up and In Vitro Embryo Production

Ovum pick-up was performed 40 h after the last FSH administration using a transvaginal ultrasound-guided technique as previously described (Hayden et al., 2022, 2023). Briefly, all visible follicles (>2 mm) from both ovaries were individually aspirated using an 18G short bevel needle fitted to a stainless-steel needle guide connected to a vacuum system (Cook Vacuum Pump; Cook Medical LLC, Bloomington, IN) set at 100 mmHg. Follicular contents of each heifer were aspirated and pooled into a single 50 mL conical tube containing Dulbecco's phosphate buffer solution supplemented with 0.4% BSA, 25 mg/L kanamycin sulfate, and 5 IU/mL sodium heparin.

Retrieval of COC from aspirates was performed using an oocyte aspiration dish with a built-in 75- μ m filter (Professional Embryo Transfer Supply, Inc., Canton, TX). Retrieved COC were evaluated using a stereomicroscope (SZ61, Olympus, Center Valley, PA) and classified into viable and nonviable following the criteria described by Hayden et al. (2022). Viable COC from each heifer were placed into in vitro maturation transport medium in a portable incubator (LabMix, WTA - Watanabe Tecnologia Aplicada, College Station, TX) and matured during transport to the location-specific laboratory. In vitro embryo production was performed at each location using proprietary standard procedures and frozen-thawed semen from 2 sires with proven fertility, one for each experiment. Cleavage and blastocyst percentages were recorded 3 and 7 d after in vitro fertilization, respectively. Embryo developmental stage and quality grade were determined 7 d after in vitro fertilization following International Embryo Technology Society guidelines (Barfield and Demetrio, 2023).

Blood Sample Collection and AMH Assessment

Blood samples for AMH evaluation were collected at a random stage of the follicular wave before the onset of the experiment. Using a 20-gauge $\times$ 1-inch (2.54 cm) needle (Vacuette Multiple Use Drawing Needle, Greiner Bio-One, Kremsmunster, Austria), blood samples were drawn by coccygeal venipuncture into 9 mL evacuated

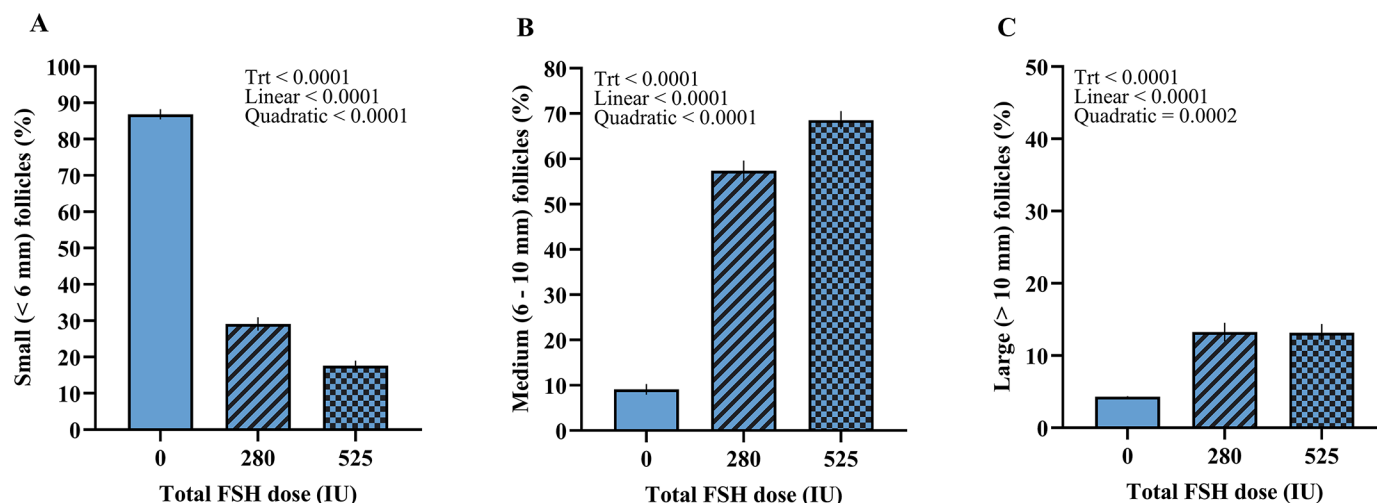


Figure 2. Percentage of small (A), medium (B), and large follicles (C) for high-AMH pregnant Holstein heifers administered 0, 280, or 525 IU of FSH before ovum pick-up (experiment 1). Error bars represent SEM. Trt = treatment.

serum tubes (Vacuette tube, Greiner Bio-One, Monroe, NC) and immediately placed on ice. Samples were then centrifuged at $1,300 \times g$ and 5°C for 13 min. The serum fraction was transferred into microcentrifuge tubes and stored at -20°C . Circulating AMH concentration was determined using an ELISA (Bovine AMH assay, Ansh Labs, Webster, TX). Intra- and interassay coefficients of variation were 1.6% and 6.2%, respectively.

Data Arrangement and Statistical Analyses

The following outcomes were analyzed for each heifer in both experiments 1 and 2: total number of follicles; number and percentage of small (<6 mm), medium (6 to 10 mm), and large (>10 mm) follicles; total number of COC; number of viable COC; number of cleaved embryos; number of blastocysts; COC retrieval percentage (total COC/total number of follicles); viable COC percentage (number of viable COC/total number of COC); cleavage percentage (total number of cleaved embryos/number of viable COC); blastocyst percentage (number of blastocysts/number of viable COC); percentage of blastocysts classified as grade 1 (number of grade 1 blastocysts/number of blastocysts); and percentage of heifers yielding ≥ 2 blastocysts.

All data were analyzed using the GLIMMIX procedure of SAS software, version 9.4 (SAS Institute Inc., Cary, NC). Generalized linear mixed models were fitted to count and binomial outcomes using the Laplace approximation method. Count data were analyzed assuming a Poisson or negative binomial distribution and using the log link function, whereas binomial outcomes were analyzed assuming a binomial distribution and using the logit link function. The linear predictor included the fixed effects

of treatment, period, and treatment sequence, and the random effects of cohort, technician performing OPU, and heifer within treatment sequence as an overarching blocking structure. The adequacy of model fit was evaluated using the maximum likelihood-based fit statistic Pearson chi-squared/df. Orthogonal polynomial contrasts were used to analyze the linear and quadratic effects of treatment on the response variables. Considering the unequal spacing between FSH doses evaluated, orthogonal polynomial coefficients were adjusted using the interactive matrix language procedure (PROC IML). A significant difference was considered when $P \leq 0.05$, whereas differences between $P > 0.05$ and $P \leq 0.10$ were considered a tendency. Data are presented as $\text{LSM} \pm \text{SEM}$.

RESULTS

Experiment 1

Quadratic effects of treatment were observed for the number ($P \leq 0.02$; Table 1) and percentage (Figure 2; $P \leq 0.0002$) of small, medium, and large follicles. The number and percentage of small follicles decreased as FSH dose increased, with a greater reduction occurring between 0 and 280 IU than between 280 and 525 IU. Conversely, the number and percentage of both medium- and large-sized follicles increased as FSH dose increased, with this increase being greater between 0 and 280 IU than between 280 and 525 IU. Consequently, the percentage of medium- to large-sized follicles (≥ 6 mm), indicative of the overall superstimulatory response, increased as FSH dose increased, albeit the increase was greater between 0 and 280 IU (0 IU: $13.2\% \pm 1.4\%$; 280 IU: $70.9\% \pm 1.8\%$; 525 IU: $82.4\% \pm 1.4\%$; $P < 0.0001$).

Table 1. Superstimulatory response and cumulus oocyte complex (COC) retrieval from high-AMH pregnant Holstein heifers administered different doses of FSH before ovum pick-up and in vitro embryo production (experiment 1; mean $\pm$ SEM)

Item	FSH dose			P-value		
	0 IU (n = 31)	280 IU (n = 31)	525 IU (n = 31)	Treatment	Linear	Quadratic
Ovarian follicles						
Small (<6 mm)	18.8 $\pm$ 1.2	7.9 $\pm$ 0.7	5.3 $\pm$ 0.5	<0.0001	<0.0001	0.02
Medium (6–10 mm)	2.1 $\pm$ 0.3	15.8 $\pm$ 1.6	20.8 $\pm$ 2.0	<0.0001	<0.0001	<0.0001
Large (>10 mm)	0.9 $\pm$ 0.2	3.7 $\pm$ 0.4	4.0 $\pm$ 0.4	<0.0001	<0.0001	<0.0001
Total	21.8 $\pm$ 1.7	27.6 $\pm$ 2.1	30.6 $\pm$ 2.3	<0.0001	<0.0001	0.20
COC retrieval						
Total COC	12.5 $\pm$ 1.7	16.0 $\pm$ 2.1	18.4 $\pm$ 2.4	<0.0001	<0.0001	0.49
COC retrieval ¹ (%)	58.1 $\pm$ 4.5	59.9 $\pm$ 4.4	61.9 $\pm$ 4.2	0.31	0.13	0.91
Viable COC	10.9 $\pm$ 1.5	14.3 $\pm$ 2.0	16.3 $\pm$ 2.2	<0.0001	<0.0001	0.34
Viable COC ² (%)	89.5 $\pm$ 1.9	91.0 $\pm$ 1.5	88.9 $\pm$ 1.7	0.51	0.81	0.28

¹Number of COC retrieved/number of total follicles.²Number of viable COC/number of total COC retrieved.

The total number of follicles as well as the total number of retrieved COC increased linearly ($P < 0.0001$) with increasing FSH dose, with no effect of treatment on COC retrieval efficacy ($P = 0.31$; Table 1). Additionally, the number of viable COC increased linearly ($P < 0.0001$) with increasing FSH dose, whereas there was no evidence ($P = 0.51$) for an effect of treatment on the percentage of viable COC (Table 1). Both cleavage and blastocyst percentage increased linearly as FSH dose increased, leading to a corresponding linear increase in the number of cleaved embryos (0 IU: 7.2 $\pm$ 1.2; 280 IU: 10.1 $\pm$ 1.7; 525 IU: 12.9 $\pm$ 2.1; $P < 0.0001$) and blastocysts ($P < 0.0001$; Figure 3). Conversely, there was no evidence ($P = 0.48$) for an FSH dose effect on the percentage of blastocysts classified as grade 1 (0 IU: 70.2% $\pm$ 6.4%; 280 IU: 69.2% $\pm$ 5.5%; 525 IU: 75.2% $\pm$ 4.6%). Finally, the percentage of heifers yielding ≥ 2 blastocysts per OPU session increased linearly with increasing FSH dose ($P = 0.02$; Figure 3).

Experiment 2

Complete data for both superstimulatory response and embryo production were unavailable for one treatment period of one heifer. Furthermore, data for embryo production were missing for one treatment period of an additional heifer due to loss of sample integrity. The number of small follicles decreased linearly ($P < 0.0001$) as FSH dose increased (Table 2). Conversely, a quadratic effect of FSH treatment ($P \leq 0.003$) was observed for the number of medium and large follicles, whereby the number of these follicles increased as FSH dose increased; however, the increase was greater between 0 and 280 IU (Table 2). There were quadratic effects of FSH dose on the percentage of small, medium, and large follicles ($P \leq 0.009$; Figure 4). The percentage of small follicles

decreased, whereas the percentage of medium and large follicles increased with increments in FSH dose, however, these changes were more pronounced between 0 and 280 IU than between 280 and 525 IU. As a result, the percentage of medium- to large-sized follicles (≥ 6 mm) increased as FSH dose increased, albeit the increase was greater between 0 and 280 IU (0 IU: 37.6% $\pm$ 4.3%; 280 IU: 73.6% $\pm$ 2.3%; 525 IU: 81.2% $\pm$ 1.9%; $P = 0.0007$). Similarly, there was a quadratic effect ($P = 0.05$; Table 2) of treatment on the total number of follicles, whereby the total number of follicles increased between 0 and 280 IU but not between 280 and 525 IU. The number of total and viable COC increased linearly ($P < 0.0001$) with increasing FSH dose, whereas quadratic effects ($P \leq 0.09$) of treatment were observed for COC retrieval efficacy and the percentage of viable COC (Table 2). The percentage of COC retrieved decreased from 0 to 280 IU of FSH but increased from 280 to 525 IU. Conversely, the percentage of viable COC increased between 0 and 280 IU but did not differ between 280 and 525 IU.

The number of cleaved embryos increased linearly (0 IU: 3.3 $\pm$ 0.5; 280 IU: 5.2 $\pm$ 0.8; 525 IU: 6.7 $\pm$ 1.0; $P < 0.0001$) with increasing FSH dose. There was, however, no evidence ($P = 0.61$) for an effect of FSH dose on cleavage percentage (Figure 5). Although the number of blastocysts increased linearly ($P < 0.0001$; Figure 5) with increasing FSH dose, a quadratic effect ($P = 0.05$) was observed for blastocyst percentage (Figure 5). Blastocyst percentage increased from 0 to 280 IU; however, it did not differ between 280 and 525 IU. Furthermore, the percentage of blastocysts classified as grade 1 increased with increasing FSH dosage in a linear ($P = 0.03$) dose-dependent manner (0 IU: 28.2% $\pm$ 5.6%; 280 IU: 39.6% $\pm$ 4.2%; 525 IU: 44.2% $\pm$ 3.9%). Finally, there was a tendency ($P = 0.07$; Figure 5) for a quadratic effect of treatment on the percentage of

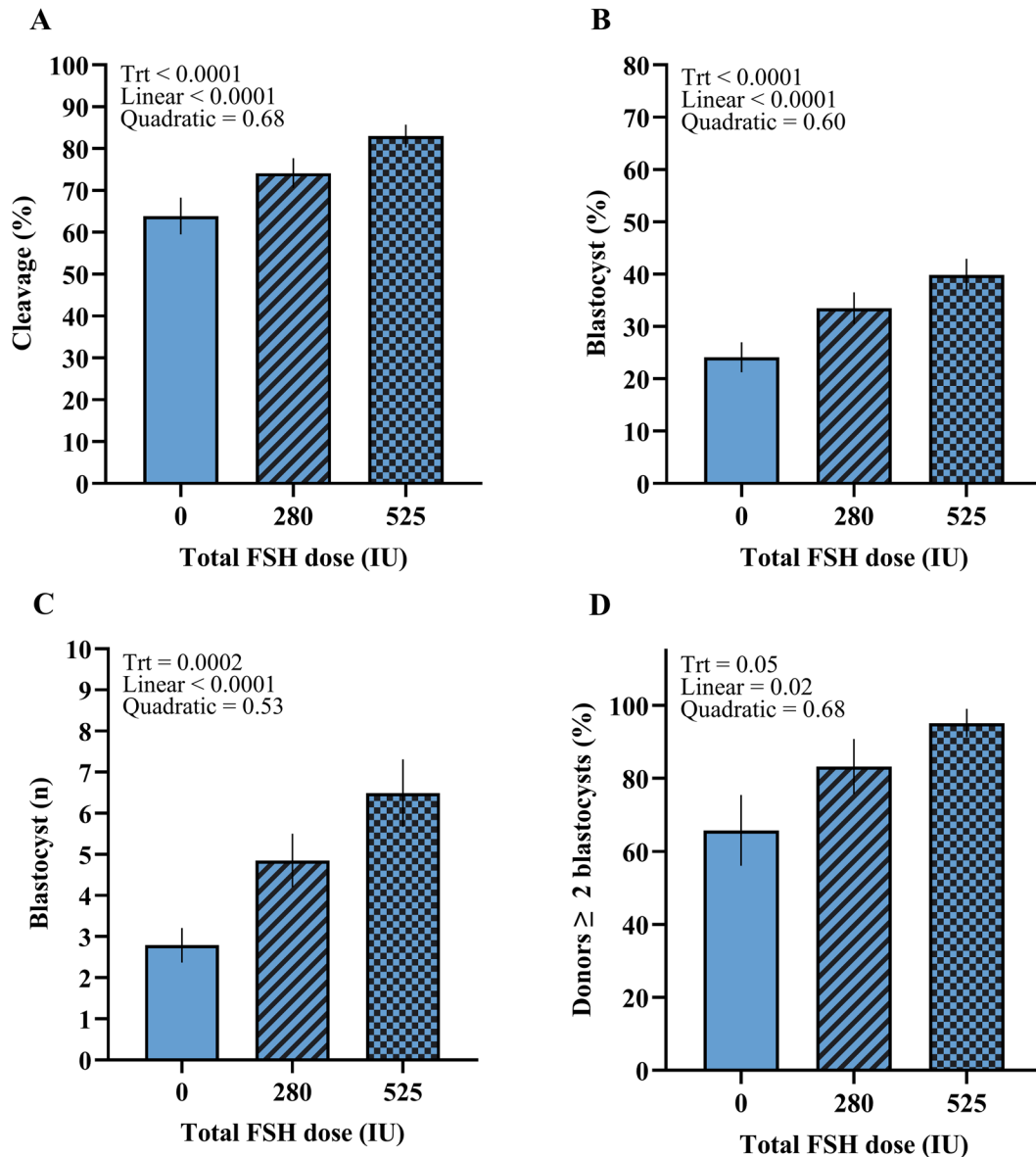


Figure 3. Cleavage percentage (A), blastocyst percentage (B), number of blastocysts (C), and percentage of donors with ≥ 2 blastocysts (D) for high-AMH pregnant Holstein heifers administered 0, 280, or 525 IU of FSH before ovum pick-up (experiment 1). Error bars represent SEM. Trt = treatment.

heifers yielding ≥ 2 blastocysts per OPU, whereby the percentage of heifers yielding ≥ 2 blastocysts increased from 0 to 280 IU but not between 280 and 525 IU.

DISCUSSION

The marked physiological differences between individuals with varying AFC/AMH (Mossa and Ireland, 2019), along with the strong association between AFC/AMH and superstimulatory response following FSH administration (Cushman et al., 1999; Singh et al., 2004; Souza et al., 2015; Motta et al., 2024), have raised the

question of whether FSH dosage should be adjusted according to AFC/AMH phenotype. The uniqueness of the present study, therefore, lies in assessing the effect of total FSH dosage on OPU/IVEP outcomes within each AMH phenotype, using a dose range commonly employed in commercial IVEP procedures. The crossover design, in which all heifers were administered every treatment, enabled each animal to serve as its own control, thereby reducing the potential for confounding effects due to individual variability. Additionally, orthogonal polynomial contrasts were used for data analysis, allowing for a more detailed understanding of the data. Nevertheless, it

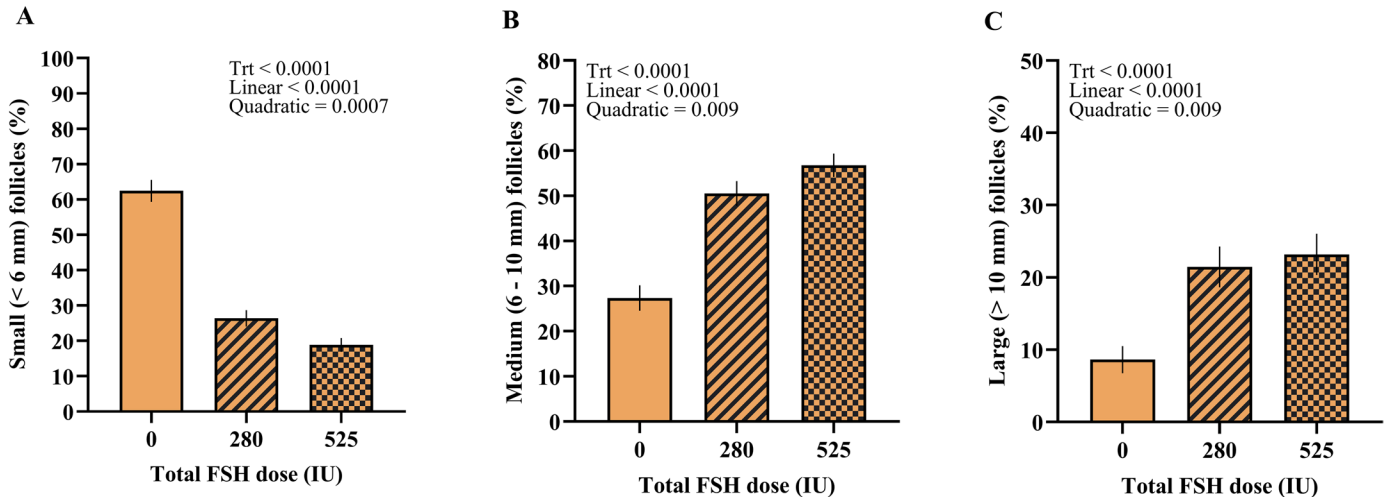


Figure 4. Percentage of small (A), medium (B), and large follicles (C) for low-AMH pregnant Holstein heifers administered 0, 280, or 525 IU of FSH before ovum pick-up (experiment 2). Error bars represent SEM. Trt = treatment.

is important to note that this study did not aim to directly compare high- and low-AMH phenotypes, as these 2 phenotypes were evaluated in separate experiments. Primary findings indicate that administration of FSH before OPU/IVEP enhances ovarian response, COC yield, oocyte developmental competence, and embryo production in both low- and high-AMH heifers, however, with variations in the type of dose-response due to AMH phenotype.

Superstimulatory response during OPU/IVEP refers to the presence of multiple medium- to large-sized ovarian follicles at the time of OPU (Hayden et al., 2022). Considering that exogenous administration of FSH rescues subordinate follicles, promoting their development into a dominant-like phenotype (Adams et al., 1993; García Guerra et al., 2015), it was hypothesized that FSH administration would increase the number and percentage of medium- and large-sized follicles. In agreement with our hypothesis, FSH affected follicle size distribution in both low- and high-AMH heifers. Specifically, FSH administration reduced the percentage of small follicles while increasing the percentage of medium and large follicles, indicating that superstimulation was achieved in both AMH phenotypes. These results are consistent with those of previous studies conducted in unselected donor females (Vieira et al., 2014; Ongaratto et al., 2020; Hayden et al., 2023). Moreover, the quadratic effect of FSH dose on both the number and percentage of medium- and large-sized follicles indicated the presence of a superstimulatory threshold consistent with previous observations in MOET (Gonzalez et al., 1990; Karl et al., 2021) and OPU/IVEP (Hayden et al., 2023). Considering the greater circulating FSH concentrations (Burns et al., 2005; Ireland et al., 2007; Jimenez-Krassel et al., 2009; Mossa et al., 2010b) and the reduced sensitivity

of granulosa cells to FSH (Scheetz et al., 2012) of low-AMH females, it was hypothesized that the superstimulatory threshold would be reached at a lower FSH dose in low-AMH heifers, and thus larger FSH doses could lead to a reduced superstimulatory response. Notably, the superstimulatory threshold seemed to be independent of AMH phenotype—at least for the doses evaluated in the present study—because a similar quadratic effect on follicle size distribution was observed in both high- and low-AMH heifers. Although somewhat unexpected, these results are consistent with the observation that follicle growth during a naturally occurring follicular wave is similar between heifers with high- and low-AFC/AMH phenotypes (Ireland et al., 2007). Furthermore, administration of excessive FSH doses to low-AFC/AMH heifers did not reduce the number of ovulatory-sized follicles (Karl et al., 2021). These results taken together indicate that the capability of follicles to grow in response to FSH administration does not appear to be affected by AMH phenotype.

Administration of FSH resulted in a significant increase in total number of follicles at OPU in both low- and high-AMH heifers, consistent with findings from multiple studies conducted in unselected donor females (Goodhand et al., 1999; Chaubal et al., 2006; De Roover et al., 2008; Vieira et al., 2016; Ongaratto et al., 2020; Hayden et al., 2023). Interestingly, however, the FSH dose-dependent effect was different in high- and low-AMH phenotype heifers. For high-AMH heifers, FSH administration increased total follicle number at OPU in a linear dose-dependent manner, in agreement with a previous report in heifers of unknown AMH phenotype (Hayden et al., 2023). In contrast, for low-AMH donors, the increase in total number of follicles as a result of

Table 2. Superstimulatory response and cumulus oocyte complex (COC) retrieval from low-AMH pregnant Holstein heifers administered different doses of FSH before ovum pick-up and in vitro embryo production (experiment 2; mean $\pm$ SEM)

Item	FSH dose			P-value		
	0 IU (n = 30)	280 IU (n = 31)	525 IU (n = 31)	Treatment	Linear	Quadratic
Ovarian follicles						
Small (<6 mm)	5.4 $\pm$ 0.9	3.5 $\pm$ 0.6	2.9 $\pm$ 0.5	<0.0001	<0.0001	0.34
Medium (6–10 mm)	2.5 $\pm$ 0.3	6.9 $\pm$ 0.7	8.7 $\pm$ 0.9	<0.0001	<0.0001	0.0002
Large (>10 mm)	0.9 $\pm$ 0.2	2.9 $\pm$ 0.5	3.6 $\pm$ 0.5	<0.0001	<0.0001	0.003
Total	9.2 $\pm$ 1.1	13.7 $\pm$ 1.5	15.6 $\pm$ 1.7	<0.0001	<0.0001	0.05
COC retrieval						
Total COC	6.3 $\pm$ 0.9	7.9 $\pm$ 1.1	10.0 $\pm$ 1.4	<0.0001	<0.0001	0.80
COC retrieval ¹ (%)	67.2 $\pm$ 4.8	57.8 $\pm$ 4.9	64.8 $\pm$ 4.6	0.03	0.44	0.008
Viable COC	4.1 $\pm$ 0.6	6.2 $\pm$ 0.8	7.9 $\pm$ 1.0	<0.0001	<0.0001	0.47
Viable COC ² (%)	68.3 $\pm$ 3.8	81.8 $\pm$ 2.7	82.0 $\pm$ 2.4	0.001	0.0009	0.09

¹Number of COC retrieved/number of total follicles.²Number of viable COC/number of total COC retrieved.

FSH administration was quadratic such that the increase in total follicles was greater (~50% vs. ~14%) between 0 and 280 IU of FSH than between 280 and 525 IU, suggesting that maximal response was nearly attained. These findings are consistent with a series of recent reports focused on FSH dose during superstimulation for MOET in low-AFC female donors (Karl et al., 2021; Lollato et al., 2022). For instance, increasing the dose of the same commercial FSH preparation as the one used in the present study from 560 to 1,680 IU did not increase the number of follicles at the end of the superstimulatory regimen in low-AFC heifers (Karl et al., 2021). Moreover, doubling the dosage of another commercial porcine FSH extract (i.e., Pluset) from 150 to 300 IU improved superovulatory response (i.e., CL number) in high- but not low-AFC *Bos indicus* female donors (Lollato et al., 2022). Together, these findings suggest that maximal ovarian response may be attained at a lower FSH dosage in low- than high-AFC/AMH female donors. Nevertheless, it is noteworthy that for high-AFC/AMH females, maximal ovarian response does not appear to be attained with current FSH dosages, and assessment of the effect of greater dosages in high-AFC/AMH females is warranted.

The efficacy of COC retrieval has been shown to decrease as follicle size increases in nonstimulated donors (Seneda et al., 2001). Therefore, it could be hypothesized that FSH superstimulation before OPU would reduce retrieval efficiency by increasing follicle size. Results from multiple studies, however, indicate that FSH administration does not affect retrieval efficacy (De Roover et al., 2008; Vieira et al., 2016; Hayden et al., 2023; García-Guerra et al., 2025). In the present study, FSH dosage was without effect on COC retrieval efficacy for high-AMH heifers. Consequently, the FSH dose-dependent linear increase in total number of follicles for high-AMH heifers led to a linear dose-dependent increase in the number of

COC retrieved. These results are consistent with those of previous studies where ovarian superstimulation resulted in an increase in both total follicles and COC (Chaubal et al., 2006; De Roover et al., 2008; Vieira et al., 2016; Hayden et al., 2023). For the low-AMH heifers, however, a quadratic FSH dose effect was observed, where COC retrieval efficacy was less following administration of 280 IU compared with 0 or 525 IU. The reason for this unexpected reduction when administering 280 IU of FSH is unclear, however, it does not appear to be associated with greater follicle size given the previously mentioned effects of FSH dose on follicle size distribution. Nevertheless, the lesser retrieval efficacy at the 280 IU dosage directly affected the number of COC retrieved. Although maximal total number of follicles in low-AMH heifers appeared to be reached at 280 IU FSH dosage, the reduction in COC retrieval efficacy at this dosage led to a linear dose-dependent increase in the total number of COC.

Retrieval of high-quality oocytes with adequate developmental competence is regarded as one of the most important factors in OPU/IVEP systems (Lonergan and Fair, 2016). Independently of AFC/AMH phenotype, oocyte developmental competence has been shown to increase as follicle size increases and appears to be associated with the acquisition of dominance (reviewed by Motta et al., 2024). Considering that ovarian superstimulation increases the proportion of medium- to large-sized follicles (Sarwar et al., 2020), it was hypothesized that oocyte developmental competence would be enhanced by FSH administration. In agreement with our hypothesis and with previous studies (Goodhand et al., 1999; De Roover et al., 2008; Vieira et al., 2014; Hayden et al., 2023; García-Guerra et al., 2025), blastocyst percentage increased when superstimulation was used in both low- and high-AMH phenotypes. These observations provide additional support for the capability of FSH to enhance

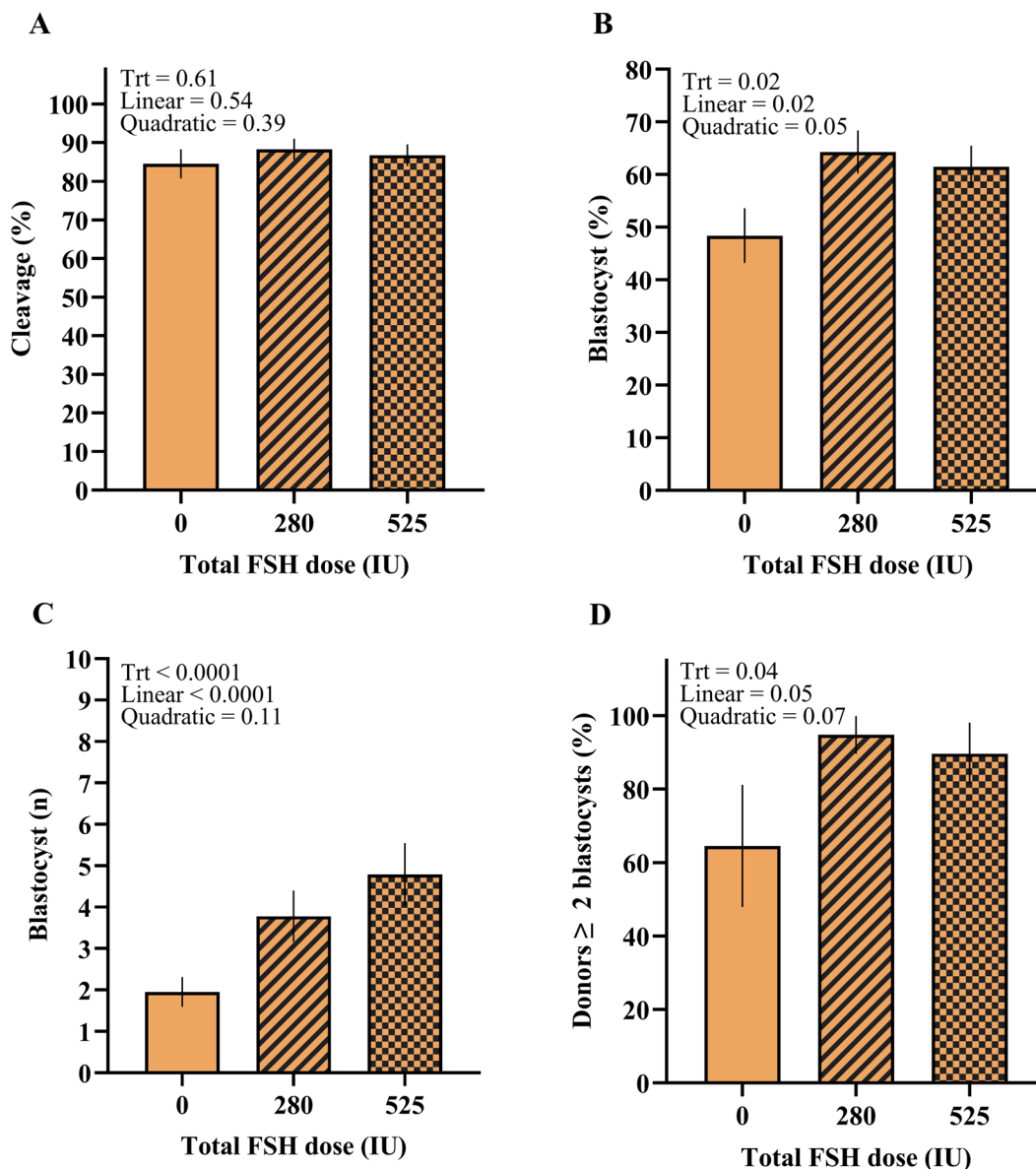


Figure 5. Cleavage percentage (A), blastocyst percentage (B), number of blastocysts (C), and percentage of donors with ≥ 2 blastocysts (D) for low-AMH pregnant Holstein heifers administered 0, 280, or 525 IU of FSH before ovum pick-up (experiment 2). Error bars represent SEM. Trt = treatment.

oocyte developmental competence and thus aid in improving IVEP efficacy.

Whether oocyte quality/developmental competence differs between low- and high-AFC/AMH phenotype females remains a highly contentious subject. Unexpectedly, in the present study, blastocyst rates were greater for low- than high-AMH heifers, which could suggest enhanced oocyte competence in low-AMH females. However, it is important to emphasize that this study was not designed to directly compare between AMH phenotypes, as these groups were evaluated in separate experiments using different sires. Therefore, these ap-

parent differences may be influenced, at least in part, by these experimental factors. Furthermore, in studies specifically designed to compare low- and high-AFC/AMH females, developmental competence has generally been reported as similar between phenotypes (Ireland et al., 2007; Nagai et al., 2016; Rosa et al., 2018; Fagundes Faria et al., 2021), with occasional positive effects favoring high-AFC/AMH females. Results from some studies suggest that oocyte quality in nonstimulated females with low AFC/AMH may be compromised as evidenced by reduced oocyte fertilizability (Nagai et al., 2015, 2016) and blastocyst rates in low- compared with

high-AFC females (Tessaro et al., 2011; Santos et al., 2016; Garcia et al., 2020). Similarly, a significant positive association between circulating AMH concentration and oocyte developmental competence has been recently reported in pubertal and pregnant superstimulated heifers (Motta et al., 2024). The precise mechanism by which AFC/AMH phenotype may affect oocyte developmental competence remains to be elucidated; however, low-AFC females were reported to have: 1) greater expression of cathepsins (*CTSB* and *CTSS*) in their cumulus cells (Ireland et al., 2009); 2) reduced oocyte mitochondrial activity (Nagai et al., 2016); and 3) follicular fluid with greater estradiol concentration (Ireland et al., 2009), all of which have been suggested to impair developmental competence (Beker-van Woudenberg et al., 2004; Bettegowda et al., 2008; Huang et al., 2016).

Low-AFC/AMH females respond poorly to FSH stimulation during assisted reproductive technologies (Cushman et al., 1999; Taneja et al., 2000; Singh et al., 2004; Ireland et al., 2007); have chronically elevated FSH secretion (Burns et al., 2005; Ireland et al., 2007, 2009; Mossa et al., 2010b); and follicular granulosa cells with apparently fewer FSHR and diminished FSH responsiveness (Scheetz et al., 2012). Furthermore, results from a recent study suggest that administration of an excessive FSH dose (i.e., 1,680 IU) to low-AFC heifers compromised oocyte developmental competence, as evidenced by a greater proportion of expanded COC, which prematurely resumed meiosis and had poor response to IVEP (Karl et al., 2023). Although the precise mechanism by which excessive FSH compromises COC quality in low-AFC cattle remains unclear, transcriptomic analysis of cumulus cells revealed altered expression of genes involved in cumulus cell expansion and the regulation of oocyte maturation (Clark et al., 2022b; Karl et al., 2023). Therefore, it was hypothesized that maximal oocyte developmental competence would be attained at a lower FSH dosage in low- compared with high-AMH heifers. Although the percentage of viable COC was not affected by FSH dose, both cleavage and blastocyst percentage increased in a linear dose-dependent manner for high-AMH heifers. Conversely, in low-AMH heifers, FSH dosage was without effect on cleavage percentage, whereas the maximal percentage of viable COC and blastocyst percentage was achieved at 280 IU and plateaued thereafter, supporting the stated hypothesis. Interestingly, administration of a high-FSH dosage in low-AMH heifers, in the present study, did not result in a reduction in oocyte developmental competence as observed previously (Karl et al., 2023), likely because even the greatest FSH dose used herein was 3.2-fold lower than the excessive dose tested in the cited study. Collectively, these results suggest that the optimal FSH dosage required to maximize oocyte

developmental competence is less for low- than high-AMH phenotype females.

The physiological mechanisms underlying the differential effect of FSH dosage on oocyte competence between low- and high-AMH phenotype heifers remain unclear. Enhanced oocyte competence following ovarian superstimulation has been linked to greater ability of cumulus cells to expand during maturation (Sugimura et al., 2017), more favorable chromatin configuration (Soares et al., 2020a,b), and a cumulus cell transcriptomic profile that favors cell-to-cell communication and an anti-inflammatory response (Sugimura et al., 2017). As previously noted, follicular cells from low-AFC/AMH heifers exhibit reduced responsiveness to gonadotropins, including a blunted follicular response to FSH compared with high-AFC/AMH counterparts (Mossa et al., 2010a; Scheetz et al., 2012). This diminished responsiveness may also involve the mechanisms associated with FSH-enhanced oocyte competence and could contribute to the observed plateau in oocyte developmental potential at higher FSH doses in low-AMH heifers, reflecting a saturation point beyond which additional FSH fails to elicit further improvements. Administration of FSH is known to induce pronounced transcriptomic changes in granulosa cells (Dias et al., 2013; Landry and Sirard, 2018), cumulus cells (Sugimura et al., 2017), and the oocyte (Chu et al., 2012); however, no studies have evaluated how these responses vary between AMH phenotypes. Future research should focus on elucidating the molecular and cellular mechanisms driving these differential responses, likely including comprehensive transcriptomic profiling of the follicular compartment across AMH phenotypes and FSH dosage. Such studies will be critical for understanding how FSH modulates follicular dynamics and oocyte quality and for refining phenotype-specific ovarian stimulation protocols that optimize IVEP outcomes.

Circulating AMH has been positively correlated with the number of blastocysts produced during OPU/IVEP in both nonstimulated (Guerreiro et al., 2014) and superstimulated cattle (Motta et al., 2024, 2025). Furthermore, the beneficial effect of superstimulation before OPU/IVEP on embryo yield has been extensively reported (Chaubal et al., 2006; De Roover et al., 2008; Vieira et al., 2014; Sarwar et al., 2020), with evidence of a dose-dependent effect of FSH (Hayden et al., 2023). Although FSH administration resulted in greater blastocyst yield in both low- and high-AMH heifers, results from the present study suggest that the dose-dependence may be different between AMH phenotypes. In high-AMH heifers, blastocyst yield and the percentage of heifers producing ≥ 2 blastocysts increased linearly as FSH dose increased, driven by the dose-dependent linear increase in total follicles, total COC, and oocyte developmental competence. For low-AMH heifers, administration of

FSH before OPU also improved blastocyst yield in a dose-dependent manner, however, it is noteworthy that the increase in blastocyst yield was more pronounced between administration of 0 and 280 IU of FSH (~94%) than the increase observed between administration of 280 and 525 IU (~27%). Furthermore, the percentage of heifers producing ≥ 2 blastocysts exhibits a quadratic effect, whereby it increased between 0 and 280 IU but remained unchanged between 280 and 525 IU. These results, therefore, suggest that maximal embryo yield is attained at a lower FSH dose in low- than high-AMH heifers, probably as a consequence of the plateau on oocyte developmental competence observed with higher FSH doses. Interestingly, the FSH dose at which maximal embryo yield is attained in high-AMH heifers is likely greater than the dosages evaluated in the present study, albeit this remains to be elucidated.

CONCLUSIONS

The results from the present study support the notion that ovarian superstimulation using FSH before OPU enhances IVEP outcomes in both high- and low-AMH heifers. Evaluation of FSH dosage, however, revealed distinct patterns of dose-dependence between AMH phenotypes, which may provide the opportunity to develop targeted ovarian superstimulatory regimens. In high-AMH heifers, greater FSH dose led to a dose-dependent linear increase in the number of follicles, viable COC, and oocyte developmental competence, resulting in greater blastocyst production. In contrast, in low-AMH heifers there were marked improvements in follicle numbers, COC viability, developmental competence, and blastocyst yield at a 280 IU FSH dose, however, increasing the FSH dose beyond 280 IU led to no further increases or relatively minor ones, suggesting that maximal response was attained. Overall, these findings suggest that the FSH dosage required to maximize IVEP efficacy is less for low- than high-AMH/AFC phenotype heifers. Nevertheless, whether further increases in FSH dose beyond the doses tested in the present study would offer any additional improvement, particularly in high-AMH heifers, remains uncertain and warrants further investigation.

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Nonstandard abbreviations used: AFC = antral follicle count; AMH = anti-Müllerian hormone; COC = cumulus oocyte complexes; ET = embryo transfer; FA = follicle ablation; IVEP = in vitro embryo production; MOET = multiple ovulation and ET; OPU = ovum pick-up; trt = treatment; US = ultrasonography.

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